# Sri Sivasubramaniya Nadar College of Engineering, Chennai
**(An Autonomous Institution affiliated to Anna University)**

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|---|---|---|---|
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| Subject Code & Name | UCS2612 – Machine Learning Algorithms Laboratory | | |
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**Experiment 4: Spam Email Detection using Linear and Kernel-Based Classification Models**

## Objective

The objective of this experiment is to build and evaluate binary classification models to distinguish spam emails from legitimate (ham) emails using Logistic Regression and Support Vector Machines (SVM). The study also focuses on understanding how hyperparameter optimization influences model performance.

## Dataset Description

The experiment utilizes the **Spambase** dataset, which consists of continuous-valued attributes derived from email text statistics. Each instance is labeled as either spam or non-spam, making it suitable for supervised binary classification tasks.

**Dataset Reference:**

- Kaggle Repository: https://www.kaggle.com/datasets/somesh24/spambase

## 3. Data Preprocessing

To ensure efficient learning and fair evaluation, the following preprocessing techniques were applied.

### 3.1 Missing Value Analysis

The dataset was inspected for null or missing values across all features. Since no missing entries were found, further handling such as imputation was not required.

### 3.2 Feature Scaling

All numerical attributes were normalized using `StandardScaler`. Standardization ensures that features contribute equally to the learning process, which is especially important for distance-based and margin-based algorithms.

### 3.3 Dataset Splitting

The data was divided into training and testing sets in an 80:20 ratio. The training set was used for model fitting and hyperparameter tuning, while the test set was reserved for final performance evaluation.

## 4. Model Implementation

The classification models were implemented using the `scikit-learn` framework with appropriate hyperparameter tuning strategies.

### 4.1 Logistic Regression

Logistic Regression classifiers were trained using both `liblinear` and `saga` solvers. Experiments were conducted with $L1$ and $L2$ regularization. The regularization parameter $C$ was tuned to control model complexity and prevent overfitting.

### 4.2 Support Vector Machine

Support Vector Machine models were developed using multiple kernels including Linear, Polynomial, Radial Basis Function (RBF), and Sigmoid. Hyperparameters such as the regularization constant $C$, kernel coefficient `gamma`, and polynomial degree were optimized using cross-validation.

### 4.3 Cross-Validation Strategy

A 5-fold cross-validation approach was adopted during hyperparameter optimization to improve robustness and reduce dependency on a single data split.

## 5. Exploratory Visualizations

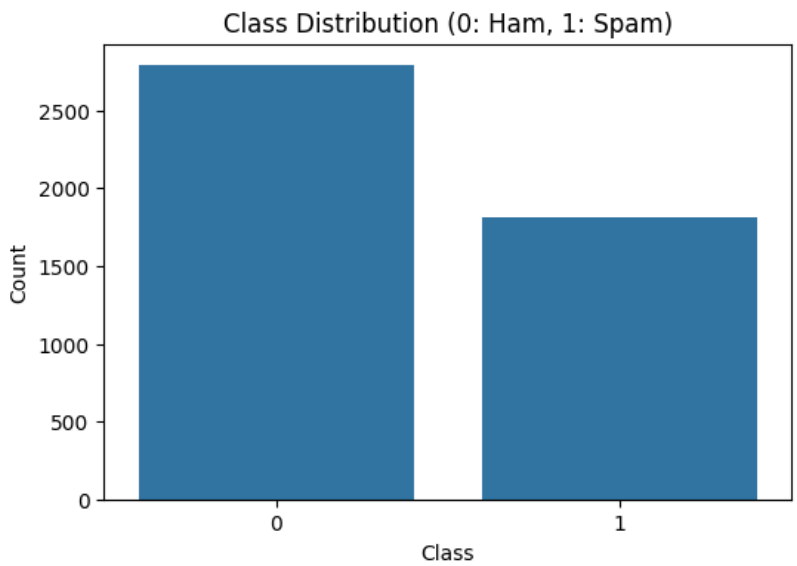


Figure 1: Distribution of Spam and Non-Spam Emails

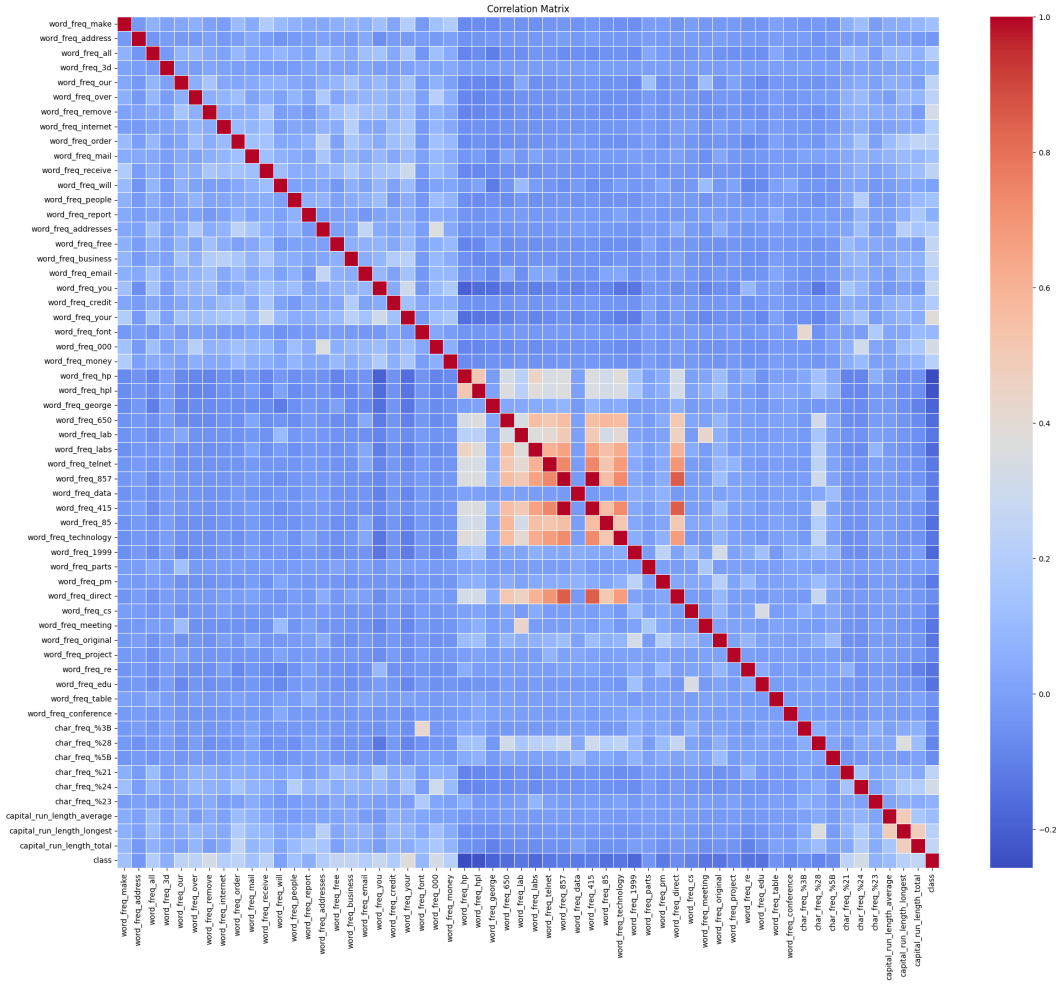


Figure 2: Feature Correlation Matrix

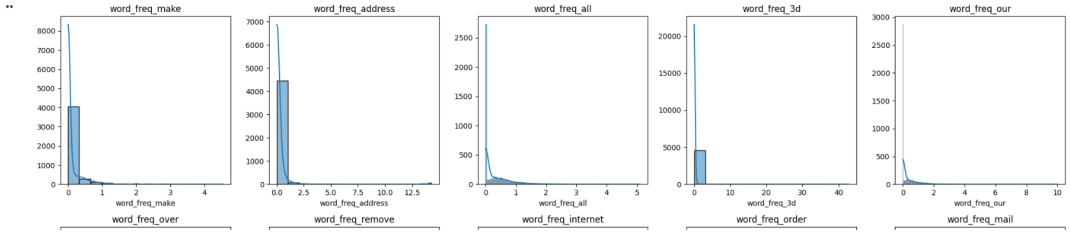


Figure 3: Feature Value Distribution Histogram

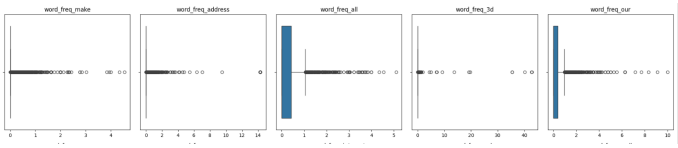


Figure 4: Boxplot Representation of Features

## Hyperparameter Optimization Results

| Model | Tuning Method | Optimal Parameters | Best CV Accuracy |
|---|---|---|---|
| Logistic Regression | Grid Search | C=5, Penalty=L2, Solver=saga | 0.9216 |
| SVM | Grid Search | C=2, Gamma=scale, Kernel=RBF | 0.9312 |

## Logistic Regression Evaluation

| Metric | Score |
|---|---|
| Accuracy | 0.9084 |
| Precision | 0.9042 |
| Recall | 0.8836 |
| F1 Score | 0.8938 |
| Training Time (s) | 2.13 |

## SVM Kernel Performance Comparison

| Kernel | Accuracy | F1 Score | Training Time (s) |
|---|---|---|---|
| Linear | 0.9149 | 0.8991 | 0.91 |
| Polynomial | 0.7812 | 0.6518 | 2.26 |
| RBF | 0.9318 | 0.9174 | 1.24 |
| Sigmoid | 0.8876 | 0.8623 | 1.19 |

## 5-Fold Cross-Validation Results

| Fold | Logistic Regression | SVM |
|---|---|---|
| Fold 1 | 0.9124 | 0.9281 |
| Fold 2 | 0.9256 | 0.9364 |
| Fold 3 | 0.8912 | 0.9448 |
| Fold 4 | 0.9437 | 0.9513 |
| Fold 5 | 0.8325 | 0.8619 |
| Average | 0.9011 | 0.9245 |

## Comparative Summary

| Aspect | Logistic Regression | SVM |
|---|---|---|
| Test Accuracy | 90.84% | 93.18% |
| Computational Cost | Low | Moderate |
| Training Speed | Faster | Slower |
| Interpretability | High | Limited |

## Results and Discussion

Experimental results demonstrate that the Support Vector Machine classifier with an RBF kernel achieved superior performance compared to Logistic Regression. The SVM model attained a test accuracy of 93.18% and an F1 score of 0.9174, indicating strong discriminative capability for identifying spam emails. Logistic Regression achieved a respectable accuracy of 90.84%, but its linear decision boundary limited its ability to capture non-linear patterns present in the data.

Hyperparameter tuning significantly improved the performance of both models. The optimized Logistic Regression model used $L2$ regularization with $C = 5$, providing a balanced trade-off between bias and variance. In the case of SVM, the selected $C = 2$ and `gamma = scale` allowed the RBF kernel to form flexible decision boundaries while maintaining good generalization.

Among the SVM kernels tested, the polynomial kernel exhibited poor performance, suggesting underfitting and ineffective feature transformation. The linear kernel performed competitively, indicating partial linear separability of the dataset. However, the RBF kernel was most effective in modeling complex feature interactions.

The close alignment between cross-validation accuracy (92.45%) and test accuracy (93.18%) for SVM indicates stable learning behavior and minimal overfitting.

## Learning Outcomes

- Gained practical understanding of linear and non-linear classifiers.
- Performed systematic hyperparameter tuning using grid search.
- Compared margin-based and probabilistic classification approaches.
- Analyzed performance using multiple evaluation metrics.

## References


- Scikit-learn Documentation: Logistic Regression
- Scikit-learn Documentation: Support Vector Machines
- Scikit-learn Documentation: Hyperparameter Tuning
- Kaggle: Spambase Dataset

- UCI Machine Learning Repository: Spambase